

Welcome to the Python course

Computational Methods I

Course WBPH064-05

Lecture 1



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Prof. Jagoda Sławińska

Zernike Institute for Advanced Materials

Theory of Condensed Matter

Room 5613.0176

jagoda.slawska@rug.nl

For any queries related to the course: **ta-cm1@rug.nl**



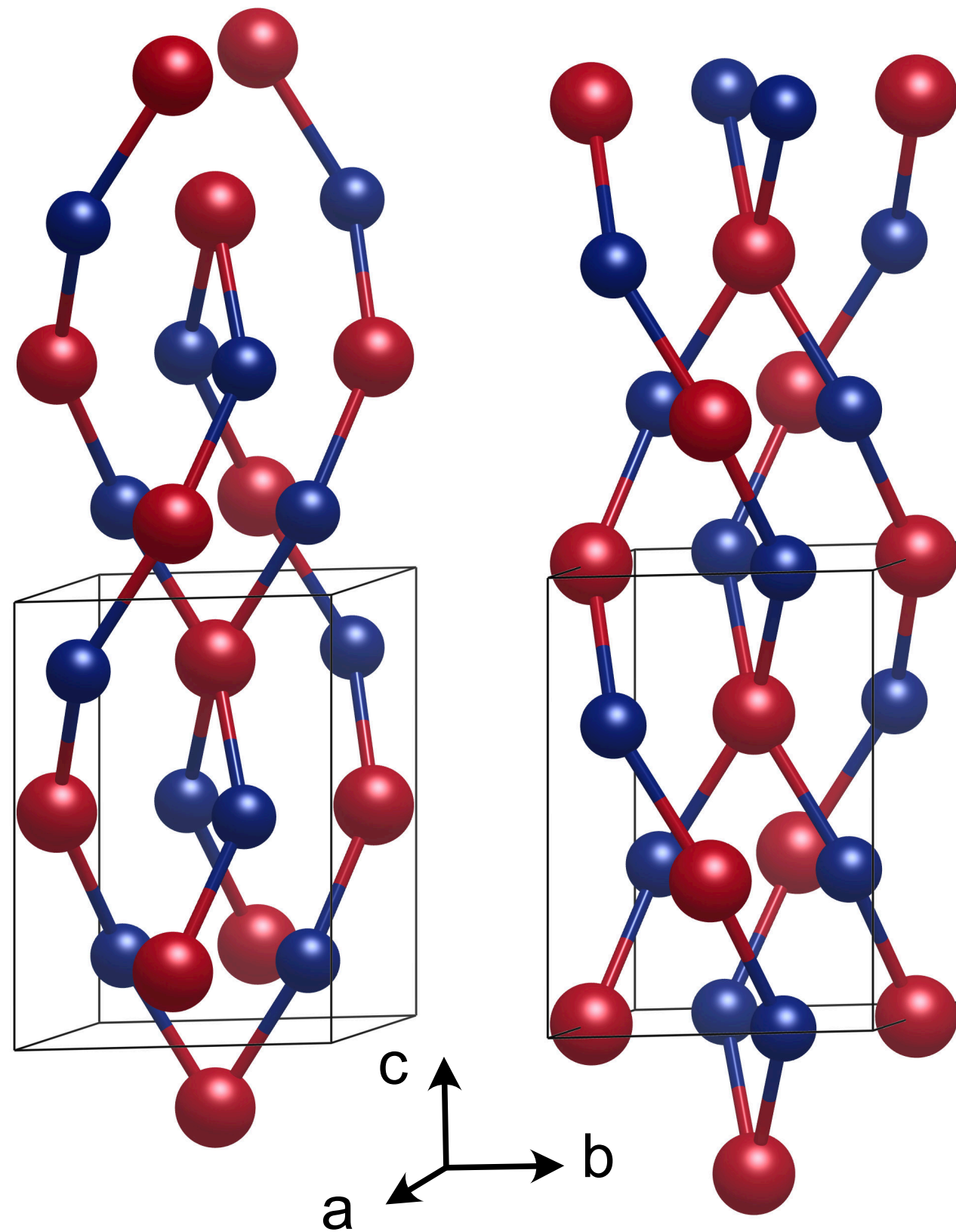
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About me

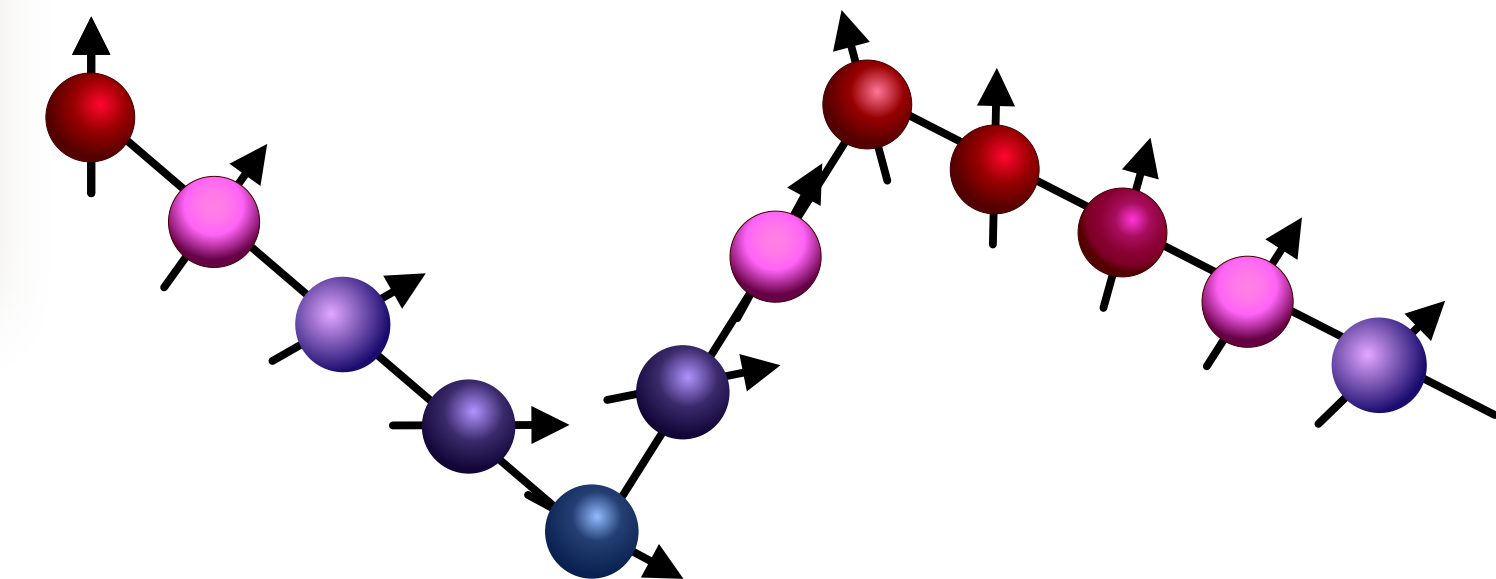
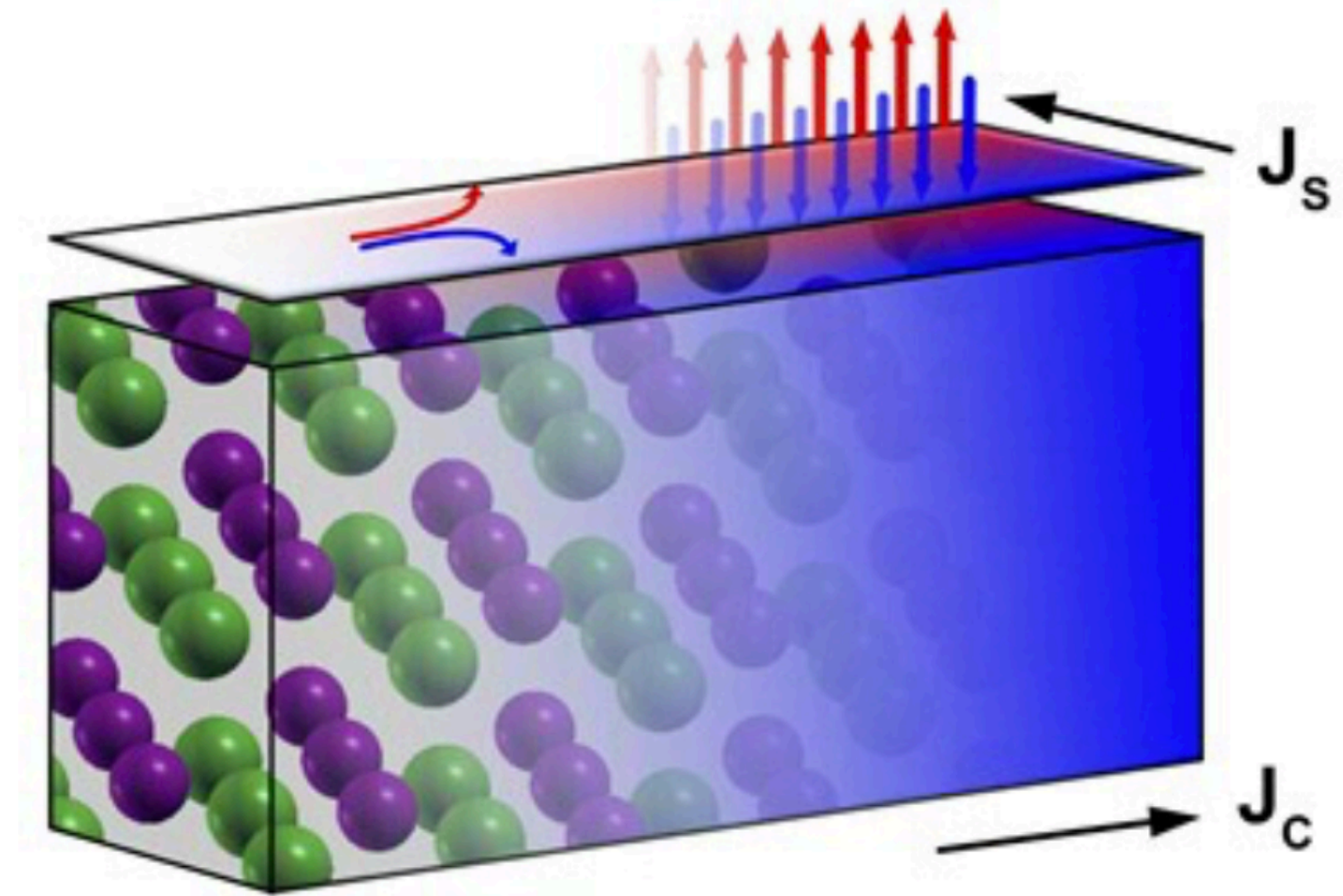


I lead a research group focused on computational materials science and doing computer simulations of materials

Our basic tool are so-called first-principles calculations

PAOFLOW

We co-develop a Python library PAOFLOW for simulations of materials properties



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Teaching assistants



Gijsbert ten Hoven



Josh Edmundson



Sergio Alvarruiz



Akram Abbasniabazeh



Federico Loi



Stephanie Gonzalez Migoni



Daniel Sade

One e-mail address: **ta-cm1@rug.nl**

Also,
Markuss Kenins
and others



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Today's outline

1. Practical information
2. Assessment
3. Why Python?
4. Notebooks
5. First codes
6. Syntax
7. Debugging



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Goals of the course

At the end of the course, you'll be able to:

- read and write simple Python programs

- manipulate (even large) sets of data

- visualize data and make high-quality plots

- apply numerical methods to solve basic physics problems

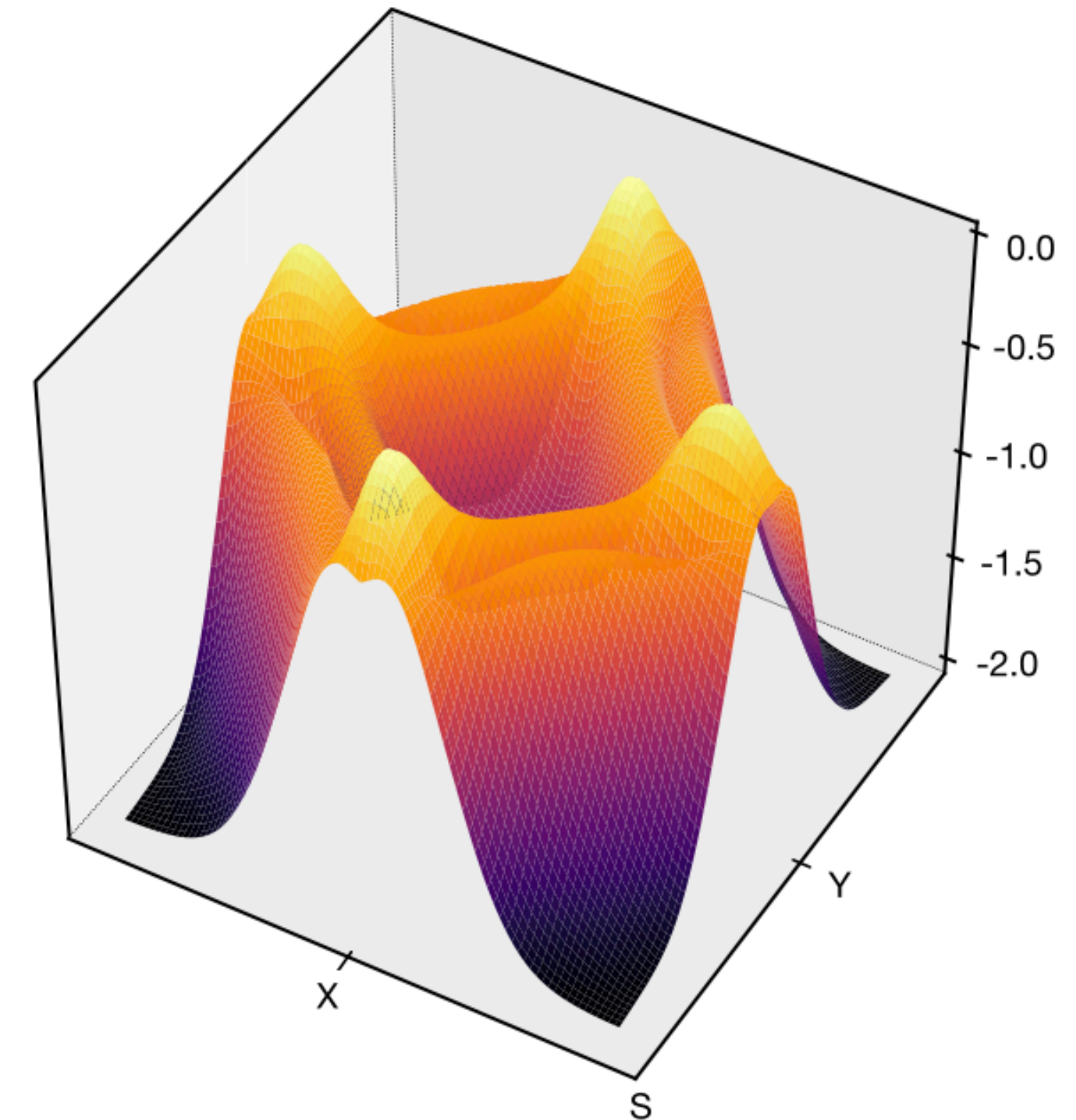
```
def gradient_and_momenta ( self,band_curvature=False ):
    '''
    Calculate the Gradient of the k-space Hamiltonian, 'Hksp'
    Requires 'Hksp'
    Populates DataController with 'dHksp'

    Arguments:
        None

    Returns:
        None
    '''
    from .defs.do_gradient import do_gradient
    from .defs.do_momentum import do_momentum
    from .defs.communication import gather_scatter
    import numpy as np

    arrays,attr = self.data_controller.data_dicts()

    try:
        snktot,nawf,_,nspin = arrays['Hksp'].shape
```



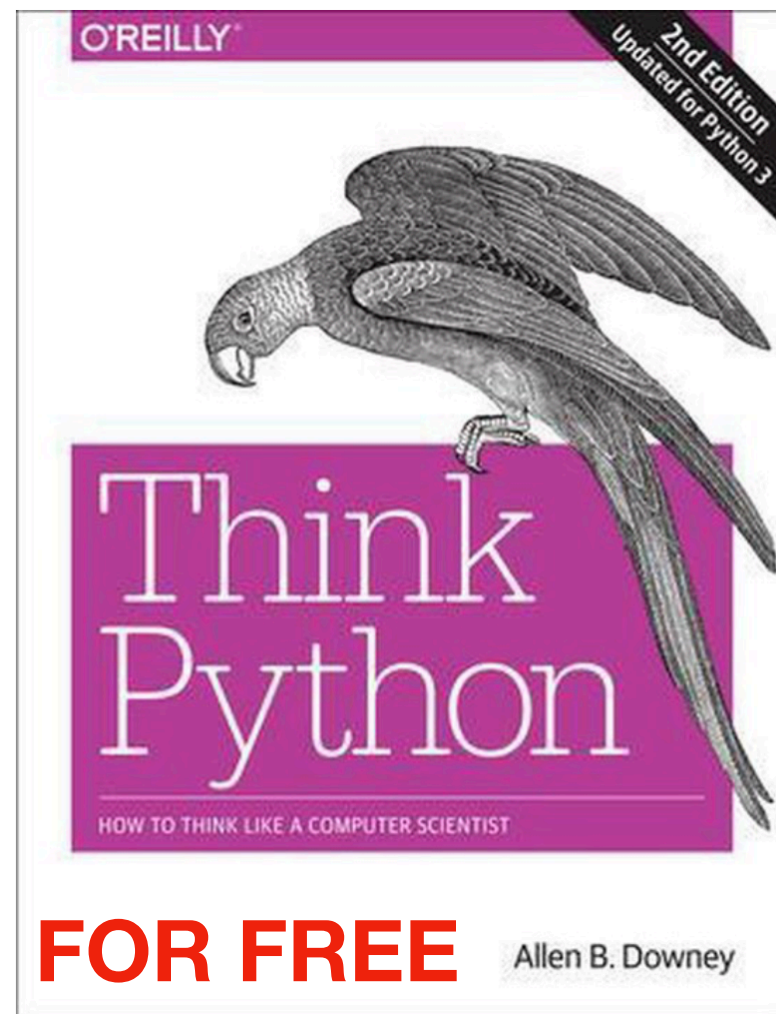
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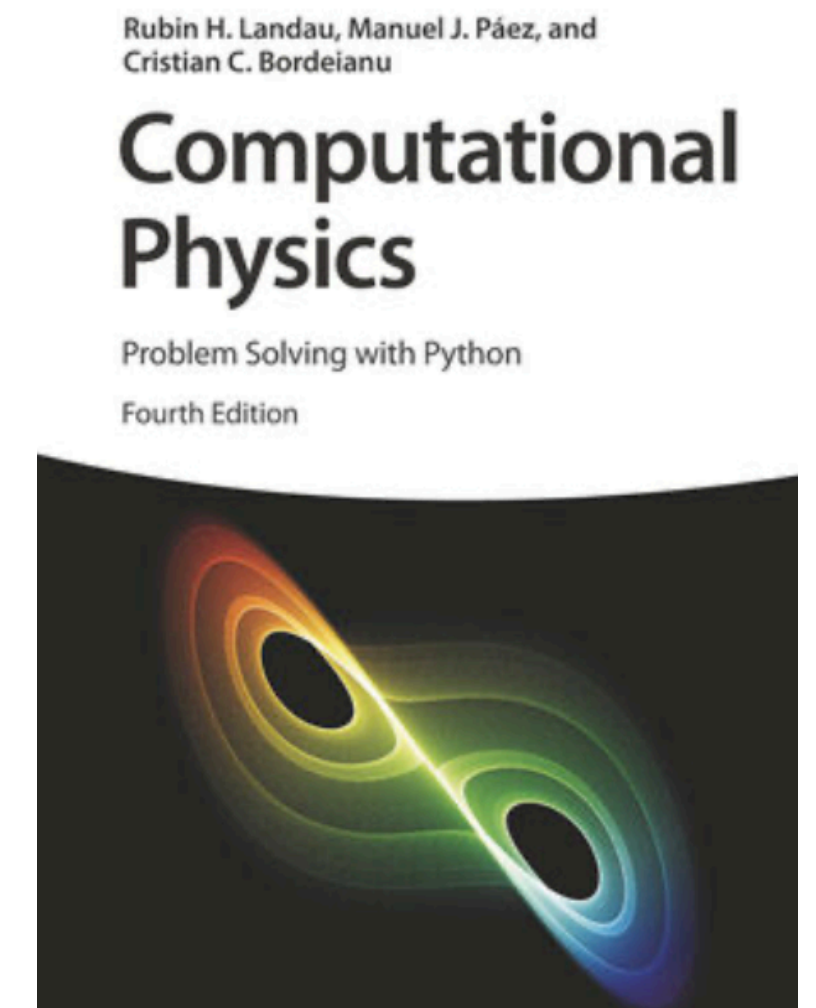
Course material



**The main course material are lecture notes available at Brightspace!
They contain a manual and tutorial exercises.**

**Lectures notes might be updated during the course;
for each update there will be a notification on Brightspace**

**There are also books, and online resources,
don't hesitate to use google and official Python manual**



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Lecture notes containing instructions and exercises to solve during tutorials
Download them now from Brightspace!

Lecture slides are a short overview of the topic of the week, will be used during lectures, correspond to chapters in the lectures notes, not mandatory

Lecture notebooks are codes created by the lecturer during the live coding at the lectures, Please go through them before starting the tutorial problems

Lectures on Mondays 17.00-19.00
except for week 5 and 6

Week 5 - lecture on Tuesday 10/03

Week 6 - no lecture on 16/03

Most of the lectures will be live coding



**Do you already have some programming experience?
Who knows what a notebook is?**



Notebooks vs. command line

- no installation on your PC is needed
- easy to share
- but: not a “real” program
- not convenient for long programs

You are not obliged to use any of these during the tutorials but please get familiar with Jupyter, as it will be your tool at the midterm and final exam



Practice makes perfect !



- go to the tutorials to get help from TA-s
- solve all exercises from the lecture notes
- solve the optional homework assignments
- discuss with your peers but do the exercises yourself

**The most important thing is to DO things by yourself.
Look things up but write them down; don't copy-paste!**



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Organization of the tutorials - sign up to one of the groups today!

Ph 9, Ph 10, Ph 11, APH 1	Akram Abbasniabazeh Stephanie Gonzalez Migoni	Tuesday 13.00 - 15.00 Wednesday 11.00 - 13.00
APH 2, APH 3, APH 4, MA 1, MA 2	Markuss Kenins Sergio Alvarruiz Campos	Wednesday 13.00 - 15.00 Thursday 13.00 - 15.00
Ph 1, Ph 2, Ph 3, Ph 4	Federico Loi Josh Edmundson	Wednesday 15.00 - 17.00 Friday 11.00 - 13.00
Ph 5, Ph 6, Ph 7, Ph 8	Gijsbert ten Hoven Daniel Sade	Wednesday 17.00 - 19.00 Thursday 11.00 - 13.00

**The TAs may occasionally swap depending on their availability
Check schedule for the last week, it is different!**



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Organization of the tutorials - mandatory presence

The presence at the tutorials is mandatory at 12 out of 16 tutorials!
If you are absent at more than 4 tutorials, the remediation track is possible in 2b
You should stay during the entire tutorial and spend time on Python

Tutorial problems	Homework problems
- basics needed to solve real problems - increase in difficulty - won't be checked - can be discussed	- combine topics of lab problems - more time-consuming - won't be checked - we can provide guidance during tutorials



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Assessment and final grade

MIDTERM
09/03/2026
18.30 - 19.30
Aletta Jacobs

+

EXAM
10/04/2026
18.15 - 20.15
Aletta Jacobs

35%

65%

No resit of the
midterm

FIVE QUESTIONS
Example: Create a
tuple of the numbers
between 0 and 100
that are divisible by 3.

You can score
0, 1 or 2 points
In each question

THREE QUESTIONS
One question similar
to the optional
homework and two
short questions

Up to 5 points for big one
2 points for each short,
1 point for coding style

The resit
possible in June



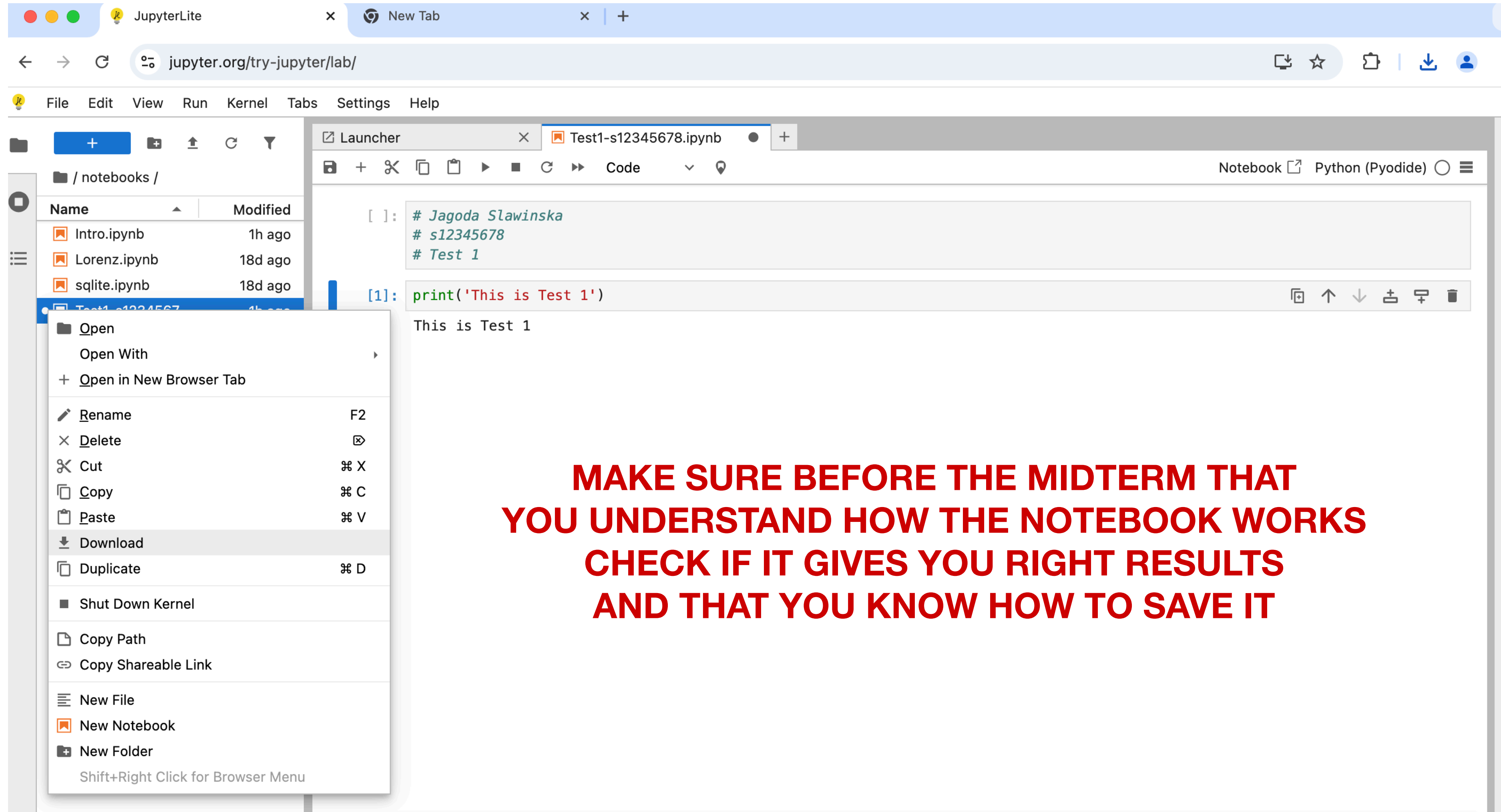
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Midterm and exam using Jupyter notebook offline



The screenshot shows the JupyterLite web interface in a browser. The address bar displays `jupyter.org/try-jupyter/lab/`. The interface includes a top menu bar with options: File, Edit, View, Run, Kernel, Tabs, Settings, and Help. On the left, a file explorer shows a directory `/ notebooks /` with files `Intro.ipynb` (modified 1h ago), `Lorenz.ipynb` (18d ago), `sqlite.ipynb` (18d ago), and `Test1-s12345678.ipynb` (1h ago). A right-click context menu is open over the selected file, listing actions such as Open, Open With, Open in New Browser Tab, Rename (F2), Delete, Cut (⌘ X), Copy (⌘ C), Paste (⌘ V), Download, Duplicate (⌘ D), Shut Down Kernel, Copy Path, Copy Shareable Link, New File, New Notebook, and New Folder. The main notebook area shows a code cell with the following content:

```
[ ]: # Jagoda Slawinska  
# s12345678  
# Test 1  
  
[1]: print('This is Test 1')
```

The output of the code cell is `This is Test 1`. The notebook is titled `Test1-s12345678.ipynb` and is using the `Python (Pyodide)` kernel.

**MAKE SURE BEFORE THE MIDTERM THAT
YOU UNDERSTAND HOW THE NOTEBOOK WORKS
CHECK IF IT GIVES YOU RIGHT RESULTS
AND THAT YOU KNOW HOW TO SAVE IT**



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Feedback for the teachers

We look forward to your feedback,
fill in the evaluation form at the end of the course,
you're also welcome to send us an email

TELL US

- what's going well, and what could improve?
- topics?
- level of difficulty
- time spent on this course



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Why Python?

- Free
- Easy to learn
- Widely used in academia and industry
 - Slower than C(++), but not in all cases
 - Much easier to learn than the above
- Useful basis for learning other programming languages
- Many good libraries (with functionalities widely used)
- Useful when studying Physics



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What is Python?



- Programming language
- Invented by Guido van Rossum
- Implementation in 1990
- The name not inspired by an animal but Monty Python



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What is Python?



- Multi-purpose (Web, GUI, Scripting, etc)
- Interpreted
- Focus on readability
- Productivity more important than speed



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It is useful for many purposes

- Automate processes
- Numerical problem solving
- Free analytical modules
- Analysis libraries
- Data visualization
- Machine learning
- ... not an exhaustive list



Repeat
Repeat
Repeat
Repeat
Repeat
Repeat



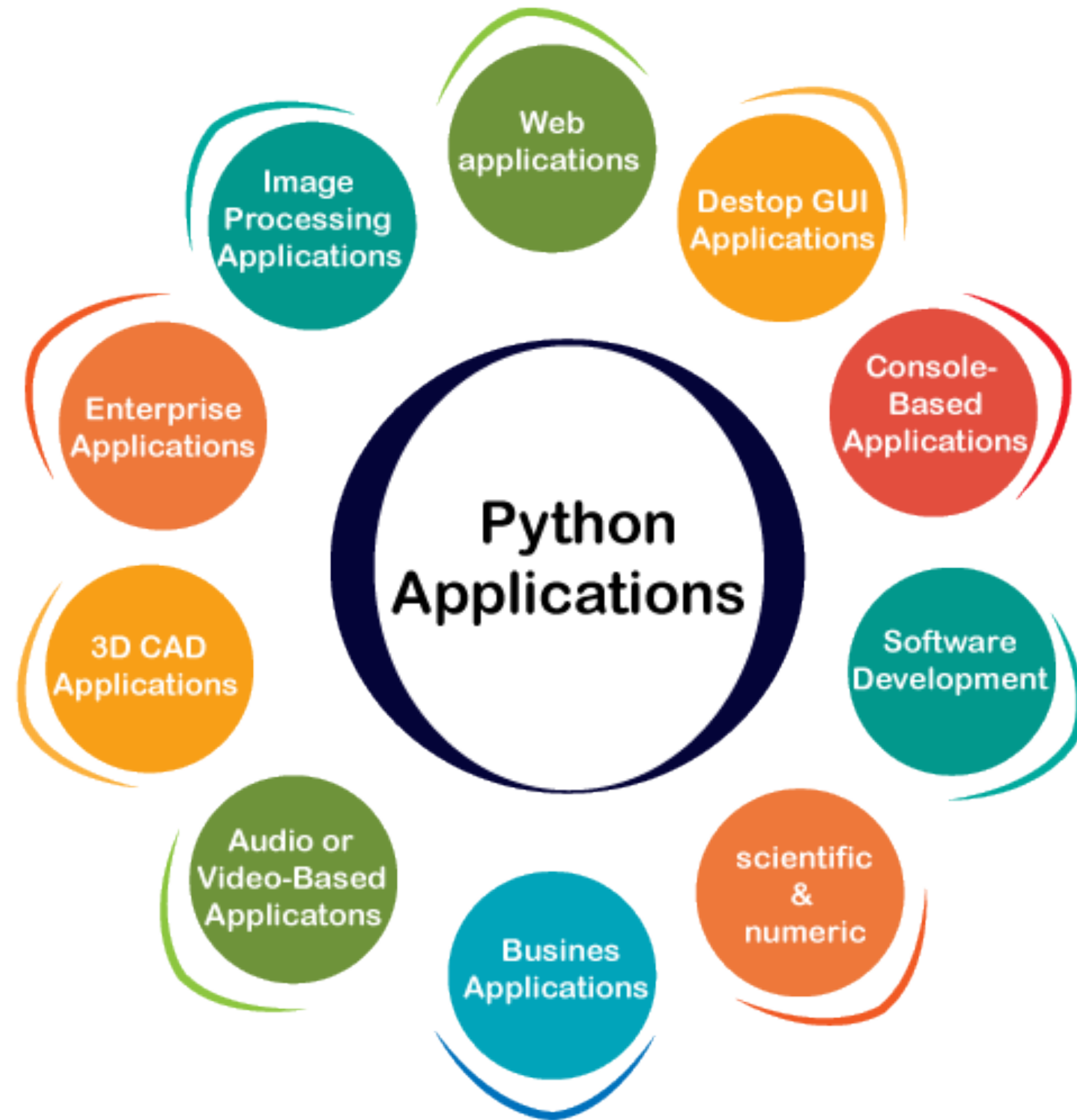
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Who uses Python?



- Universities and research institutes
- Google
- CERN
- NASA
- ABN AMRO
- Instagram
- Spotify
- Amazon
- ... many more



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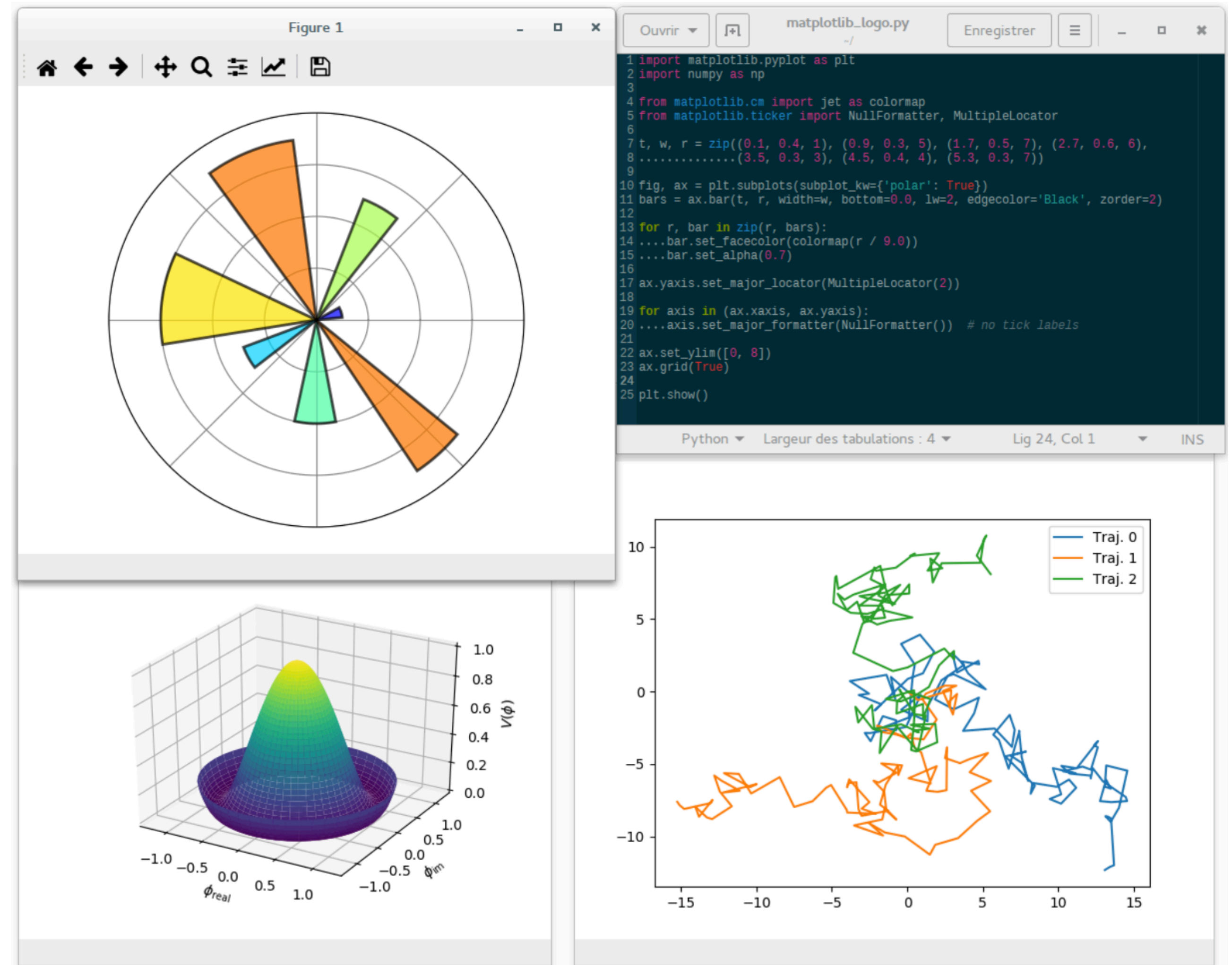
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- Created in 1989 by Guido Van Rossum
- Python 1.0 released in 1994
- Python 2.0 released in 2000
- Python 3.0 released in 2008
- You'll find many people using 2.x however:
- Python 3.x is the present...
- Programming languages develop over time!



Course planning

1. Basics
2. Loops and comments
3. Functions, modules, packages
4. Plotting
5. Data analysis
6. Numerical methods
7. Monte Carlo simulations



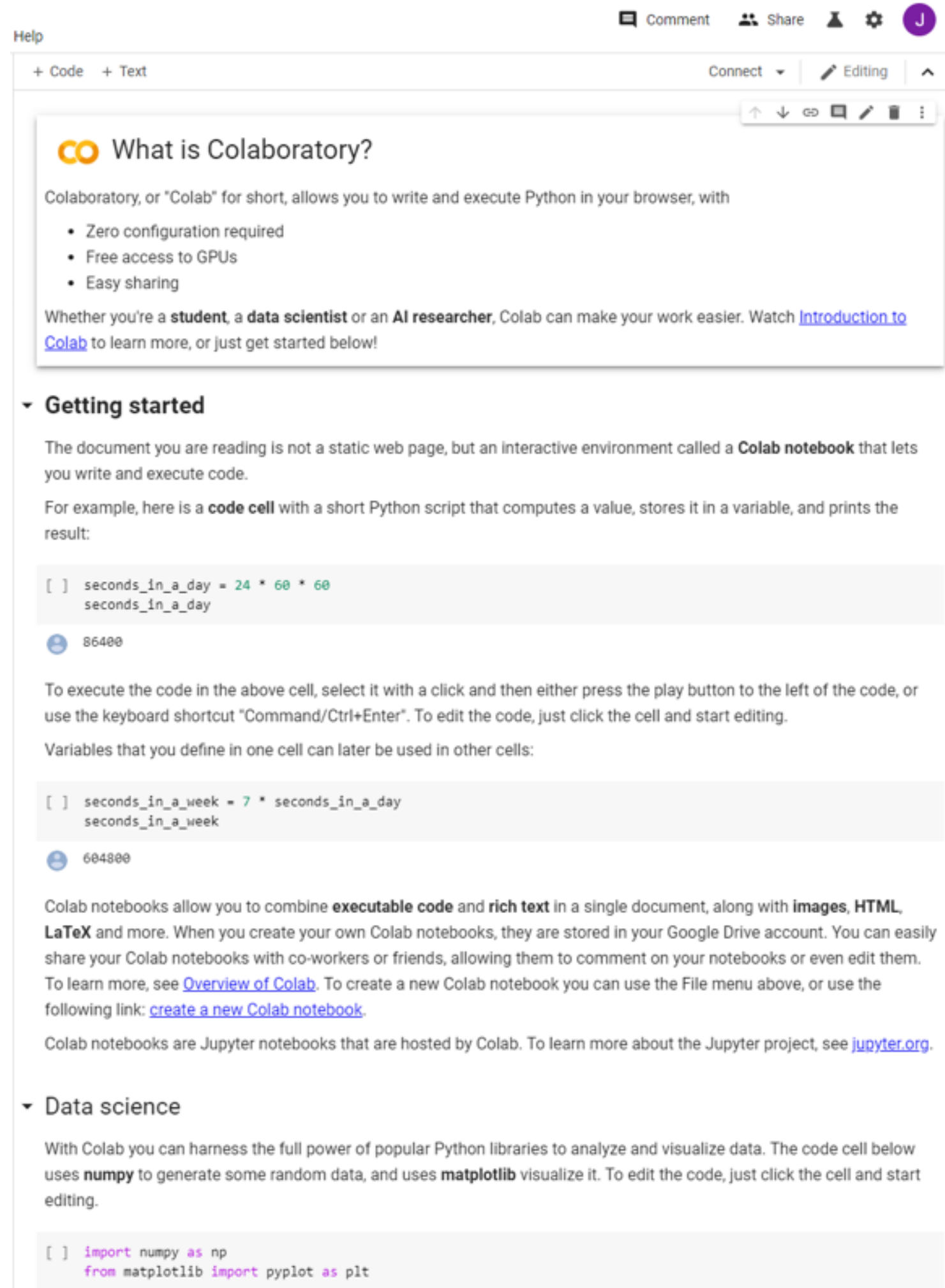
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Working with notebooks



- Editor and compiler
- Easy to share
- Executes codes
- Displays hints
- Debugging on the fly
- Easy access to plots
- You see what you're doing

Google colab



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Syntax is the set of rules that defines how a Python program should be written, and interpreted.

Python is an easy-to-read language, however good comments, explaining what a piece of code does, improves readability.

Indentation in the script is actually read by the program! Not just for readability.

Whatever comes after a “#” on a line will not be read by the computer, it’s for humans only...

```
1  #This comment will be ignored by the interpreter
2  print ("Comment Example")
3  |
```



'Hello world'

It is very easy to start coding in Python

✓
0 s



```
print('Hello world')
```

Hello world

Google colab



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Object-oriented programming

- Variable can be used directly without declaration, e.g. `a = 1` or `b = "my name"`
- Data type can change, pro and con! Be careful
- Interpret language
- Write once, run everywhere
- Garbage collection
- Scripting language
- Extract information from data set



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Object-oriented programming

The object is stored somewhere in the memory and can be called with a variable
Note: a variable in programming is not the same as a variable in a function!

```
testlist = [3, 0, 9, 2, 5]
```

variable

object (list)



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- **Integers**
- Floating-point numbers
- Complex numbers
- Strings
- Boolean
- Tuple
- List

```
[3] # integers
    a = 3
    b = 5
    c = a + b
    print(type(a))
    print(type(c))
```

```
↳ <class 'int'>
   <class 'int'>
```



- Integers
- **Floating-point numbers**
- Complex numbers
- Strings
- Boolean
- Tuple
- List

```
[5] # float
    a = 3.
    b = 7
    c = a + b
    print(type(a))
    print(type(b))
    print(type(c))
```

```
↳ <class 'float'>
   <class 'int'>
   <class 'float'>
```



- Integers
- Floating-point numbers
- **Complex numbers**
- Strings
- Boolean
- Tuple
- List

```
[10] # complex numbers
      a = 2j
      b = a ** 2
      print(b)
      print(type(b))
```

```
[-4+0j]
<class 'complex'>
```



- Integers
- Floating-point numbers
- Complex numbers
- **Strings**
- Boolean
- Tuple
- List

```
[11] # strings
      a = "This is a string"
      b = "3"
      c = 3 * b
      print(type(a))
      print(type(b))
      print(c)
      print(type(c))
```

```
↳ <class 'str'>
   <class 'str'>
   333
   <class 'str'>
```



- Integers
- Floating-point numbers
- Complex numbers
- Strings
- **Boolean**
- Tuple
- List

```
[16] # Boolean
      a = 4
      b = 2
      print(a == b)
      print(a < b)
      print(a != b)
      print(type(a == b))
```

```
↳ False
   False
   True
   <class 'bool'>
```



- Integers
- Floating-point numbers
- Complex numbers
- Strings
- Boolean
- Tuple
- **List**

```
[23] # lists
      a = ["item 0", "item 1", "item 2"]
      print(a)
      print(a[0])
      a[0] = "blabla"
      print(a)
      print(type(a))
```

```
☞ ['item 0', 'item 1', 'item 2']
   item 0
   ['blabla', 'item 1', 'item 2']
   <class 'list'>
```



- Integers
- Floating-point numbers
- Complex numbers
- Strings
- Boolean
- Tuple
- **List**

```
[16] # Boolean
      a = 4
      b = 2
      print(a == b)
      print(a < b)
      print(a != b)
      print(type(a == b))
```

```
↳ False
   False
   True
   <class 'bool'>
```



- Integers
- Floating-point numbers
- Complex numbers
- Strings
- Boolean
- **Tuple**
- List

```
[24] # tuple
      a = ("item 0", "item 1", "item 2")
      print(a)
      print(a[0])
      print(type(a))
```

```
↳ ('item 0', 'item 1', 'item 2')
   item 0
   <class 'tuple'>
```

The same as lists?



- Integers
- Floating-point numbers
- Complex numbers
- Strings
- Boolean
- **Tuple**
- List

```
a[0] = "blabla"  
print(type(a))
```

```
↳ ('item 0', 'item 1', 'item 2')
```

TypeError

Traceback (most recent call last):

<ipython-input-25-226ba3067d04> in <module>()
1 a = ("item 0", "item 1", "item 2")
2 print(a)
----> 3 a[0] = "blabla"
4 print(type(a))

TypeError: 'tuple' object does not support item assignment

Tuples are immutable objects



Variables types and naming rules

Variables can refer to any type but

- it can lead to errors
- variables will change their type if you tell them to

Rules for naming

- variable name cannot be a Python keyword
- variable name cannot contain spaces
- first character must be a letter or an underscore
- after first character may use letters, digits, or underscores
- variable names are case-sensitive



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Indentation

- Most languages don't care about indentation
- Most humans however do care
- Grouping similar things together gives a nice overview
- Python cares! Common cause of errors.... Be warned :)

```
if 1 > 2:  
    print("1 is smaller than 2")  
    print("This line of code will also be executed because of indentation")
```



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Good luck with the course ...



HAPPY CODING!



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TELL US ABOUT YOUR COURSE PROGRAMME!

Check your student mail and
take the National Student Survey!



LOST YOUR MAIL?
SCAN THE CODE!

